Symmetric Group.

Def. Let A be a non-empty set.

A Group of bijections on A;

$$S_A \overset{\text{def}}{=} \{ f; A \to A \mid f \text{ is bijection} \}$$

is called a Symmetric Group on A.

Lemma. Define S_n as Symmetric group on a set $\{1, 2, \cdots, n\}$.

If $|A| = n$, then $S_A \cong S_n$.

Proof. Since $|A| = n$, there exists a bijection

$$f; A \to \{1, 2, \cdots, n\}$$

Define a map:

$$\varphi ; \underset{A \to A}{S_A \to S_n} ; \sigma \mapsto \underset{A \to \{1,\cdots,n\}}{f} \circ \underset{A \to A}{\sigma} \circ \underset{\{1,\cdots,n\} \to A}{f^{-1}}$$

It is well defined, because Composition of bijection is bijection.

And, injectiveness is given by just Cancelation;

$$f \circ \sigma_1 \circ f^{-1} = f \circ \sigma_2 \circ f^{-1} \Rightarrow \sigma_1 = \sigma_2$$

Surjectivity is given by; for any $\alpha \in S_n$, $f^{-1} \circ \alpha \circ f$ is contained in S_A and

$$\varphi(f^{-1} \circ \alpha \circ f) = f \circ f^{-1} \circ \alpha \circ f \circ f^{-1} = \alpha.$$

Finally, Homomorphism because; for any $\sigma_1, \sigma_2 \in S_A$,

$$\varphi(\sigma_1 \circ \sigma_2) = f \circ \sigma_1 \circ \sigma_2 \circ f^{-1} = f \circ \sigma_1 \circ f^{-1} \circ f \circ \sigma_2 \circ f^{-1} = \varphi(\sigma_1) \circ \varphi(\sigma_2).$$

∎

Now, we notate all permutation group as S_n.

「Cycle Decomposition.

Let S_n be a Symmetric Group. Then, every element of S_n can be represented by

$$\sigma = \begin{pmatrix} 1 & 2 & \cdots & n \\ a_1 & a_2 & \cdots & a_n \end{pmatrix}$$ where $\{a_i\}_{i=1}^{n}$ is a sequence having range as $\{1, \cdots, n\}$.

Choose $a \in \{1, \cdots, n\}$. Construct a cycle: for the smallest positive integer $k_1 \geq 1$ s.t $\sigma^{k_1}(a) = a$,

$$(a \;\; \sigma^1(a) \;\; \sigma^2(a) \cdots \sigma^{k_1 - 1}(a))$$

Choose $b \in \{1, \cdots, n\}$ s.t $\forall\, 0 \leq j \leq i_1 - 1$, $b \neq \sigma^j(a)$. Then, similarly, we can construct:

$$(b \;\; \sigma^1(b) \;\; \sigma^2(b) \cdots \sigma^{k_2 - 1}(b))$$

Since $\{1, \cdots, n\}$ is finite set, this process will terminate in finite time.

And, the elements in a cycle are mutually distinct, because: if for some $0 \leq i < j \leq k_1 - 1$,

$$\sigma^i(a) = \sigma^j(a) \Rightarrow \sigma^0(a) = a = \sigma^{j-i}(a)$$

But this contradicts to minimality of k_1.

Moreover, these two cycles are disjoint: if for some $0 \leq r \leq k_1 - 1$, $0 \leq t \leq k_2 - 1$,

$$\sigma^r(a) = \sigma^t(b) \Rightarrow b = \sigma^{r-t}(a)$$

「Cycle decomposition.

Let S_A be a symmetric group of a set A. Let $\sigma \in S_A$ be given.

Define a relation on A as:

$$a \sim b \iff a = \sigma^i(b) \text{ for some } i \in \mathbb{Z}.$$

Then, it is an equivalent relation because

$a \sim a$ is trivial, put $i=0$.

If $a \sim b$, then $a = \sigma^i(b)$ for some $i \in \mathbb{Z}$. Take σ^{-i} to both side, then $\sigma^{-i}(a) = b$.

If $a \sim b$ and $b \sim c$, then, for some $i, j \in \mathbb{Z}$, $a = \sigma^i(b)$ and $b = \sigma^j(c)$. Now,

$$a = \sigma^i(b) = \sigma^i(\sigma^j(c)) = \sigma^{i+j}(c)$$

Therefore, we can consider a partition

$$A/\sim = \{C_a \mid a \in A\} \text{ where } C_a = \{\sigma^i(a) \mid i \in \mathbb{Z}\}.$$

If A is infinite, A can has infinitely many infinite orbit.

But, if A is finite, A must has finitely many finite orbit.

Regardless of Cardinality of A, the orbit have the following properties:

「Lemma. Let A be a set, and let $\sigma \in S_A$ be a permutation on A.

If $a \in A$ satisfies that $\sigma^n(a) = a$ for some $n \geq 0$,

then, for $p = \min\{n \geq 0 \mid \sigma^n(a) = a\}$,

$$\{a, \sigma(a), \cdots, \sigma^{p-1}(a)\} = C_a$$

and each $\sigma^i(a)$ $(0 \leq i \leq p-1)$ are mutually disjoint.

Proof. The existence of such p is given by the Well-ordering principle.

And, Since $\sigma^p(a) = a$, for any $k \in \mathbb{Z}$,

$$\sigma^k(a) = \sigma^{k \bmod p}(a).$$

Finally, to show that mutually disjointness, Suppose that for some $0 \leq i < j \leq p-1$,

$$\sigma^i(a) = \sigma^j(a)$$

Take σ^{-i} to both side, then

$$a = \sigma^{j-i}(a)$$

But it is contradiction to the minimality.

If exists, the smallest number $p \geq 1$ s.t. $\sigma^p(a) = a$ is called period of a with respect to σ.

Clearly, if C_a is finite set, then all elements have a same period.

Def. Let $\sigma \in S_n$ be a permutation.

If σ has at most one cycle having more than one element, then σ is called a cycle, and denote

$$\sigma = (a \ \sigma(a) \ \sigma^2(a) \ \cdots \ \sigma^{p-1}(a)) \text{ where } p \geq 1 \text{ is period of } a.$$

Thm. Let S_n be a Symmetric Group on n letters.

Every permutation of S_n can be represented by finite composition of disjoint cycles. Moreover, disjoint cycles are commute.

Proof. Let $\sigma \in S_n$ be given. Then, there exist finite orbits of σ;

$$C_{a_1}, \cdots, C_{a_k}.$$

For each $1 \leq i \leq k$, define;

$$\sigma_i : \{1, \cdots, n\} \to \{1, \cdots, n\} ; x \mapsto \begin{cases} \sigma(x) & \text{if } x \in C_a \\ x & \text{if } x \notin C_a \end{cases}$$

Then, clearly each σ_i's are cycles, and commute. Moreover,

$$\sigma = \sigma_1 \circ \cdots \circ \sigma_k.$$

「Transposition decomposition.

「Def. Let S_n be a Symmetric Group on n letters.
A length of 2 cycle is called a transposition in S_n.

「Thm. Every Cycles in S_n Can be represented by finite product of transpositions.

Proof. Let a Cycle $(a_1\ a_2\ \cdots\ a_k) \in S_n$ be given. Then, by computation,

$$(a_1\ a_2\ \cdots\ a_k) = (a_1\ a_k)(a_1\ a_{k-1})\cdots(a_1\ a_2)$$

Now, we can consider all permutation in S_n is a product of transpositions.

Note; transposition decomposition is not unique. for example,

$$(1\ 2\ 3) = (1\ 3)(1\ 2) = (1\ 3)(1\ 2)(1\ 3)(1\ 3)$$

「Thm. No permutation in S_n Can be represented by both even and odd product of transpositions.

Proof. Let $\sigma \in S_n$ be given. To use Contradiction, Suppose that

$$\sigma = f_1 \circ \cdots \circ f_{n_1} = g_1 \circ \cdots \circ g_{n_2}$$

where f_i, g_i are transposition, and n_1 is odd, n_2 is even number.

Consider the standard basis for $\mathbb{R}^n$,

$$\{e_1, \cdots, e_n\}$$

Clearly,

$$\det(e_1\ \cdots\ e_n) = 1.$$

And,

$$\det\left(e_{f_{n_1}(1)}\ e_{f_{n_1}(2)}\ \cdots\ e_{f_{n_1}(n)}\right) = -1$$

Because it is the change of two rows for I_n.

Inductively,

$$\det\left(e_{f_1\circ\cdots\circ f_{n_1}(1)}\ \cdots\ e_{f_1\circ\cdots\circ f_{n_1}(n)}\right) = (-1)^{n_1} = -1$$

and

$$\det\left(e_{g_1\circ\cdots\circ g_{n_2}(1)}\ \cdots\ e_{g_1\circ\cdots\circ g_{n_2}(n)}\right) = (-1)^{n_2} = 1.$$

Contradiction.

Thm. Let S_n be a Symmetric Group on n letters.

1. Let $\sigma \in S_n$ is a cycle of length m. Then

   σ^i is m-cycle if and only if $\gcd(i, m) = 1$

Proof. Denote $\sigma = (a_0\ a_1\ \cdots\ a_{m-1})$ as cycle notation. That is, $\sigma^k(a_0) = a_{k \bmod m}$.

Let $2 \leq i \leq m-1$ be given.

σ^i is m-cycle $\Leftrightarrow a_i, \sigma^i(a_i), \sigma^{2i}(a_i), \cdots, \sigma^{(m-1)i}(a_i)$ are distinct

$\Leftrightarrow a_i, a_{2i \bmod m}, a_{3i \bmod m}, \cdots, a_{mi \bmod m}$ are distinct.

$\Leftrightarrow \forall x, y \in \{1, \cdots, m\},\ x \neq y \Rightarrow xi \not\equiv yi \bmod m$

$\Leftrightarrow \gcd(i, m) = 1.$

2. $\sigma \in S_n$ has order 2 iff σ is product of commute transpositions.

Proof. Let $\sigma \in S_n$ has order 2. That is, for all $1 \leq i \leq n$, $\sigma^2(i) = i$ and $i \neq \sigma(i)$.

Now, a cycle in the cycle must be identity or transposition.

Since order of σ is not 1, it contains at least one transposition.

The converse is trivial.

3. Let p be a prime number. Then,

   $\sigma \in S_n$ has order p iff σ is product of commute p-cycles.

4. Let $\sigma \in S_n$ be a permutation with cycle decomposition

$$\sigma = \sigma_1 \circ \cdots \sigma_k.$$

Then,
$$|\sigma| = \operatorname{lcm}(|\sigma_1|, \cdots, |\sigma_k|).$$

Note: we already know that the map

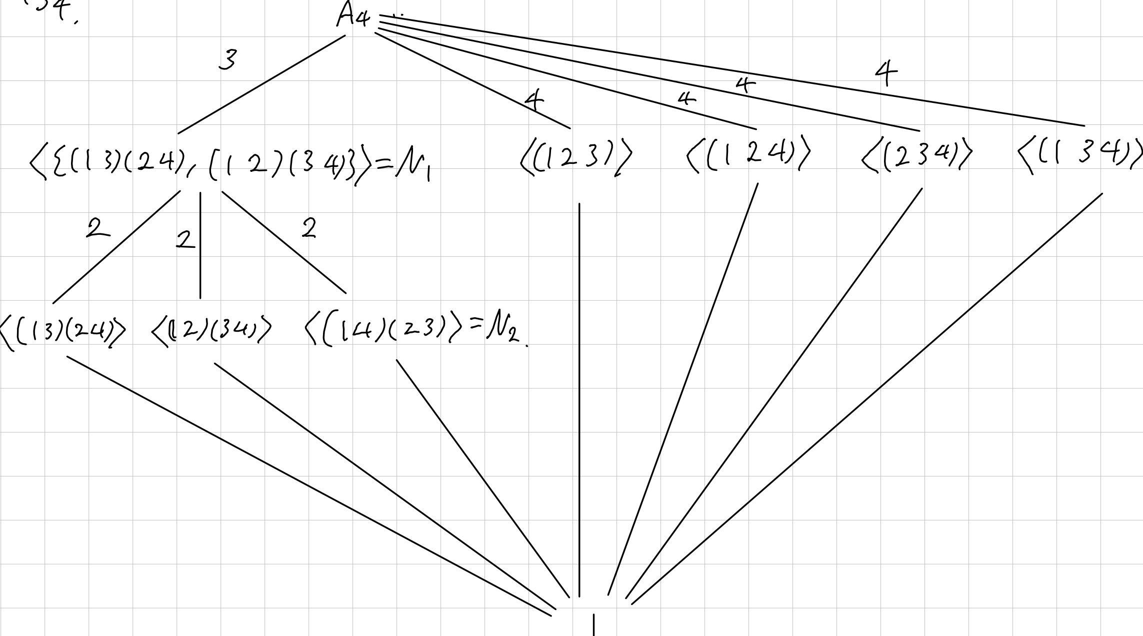
$$\mathrm{Sgn}: S_n \to \{\pm 1\}: \sigma \mapsto \begin{cases} 1 & \text{if } \sigma \text{ is even} \\ -1 & \text{if } \sigma \text{ is odd} \end{cases}$$

is well-defined.

Lemma. $\mathrm{Sgn}: S_n \to \{\pm 1\}$ is Epimorphism.

Def. Let S_n be a Symmetric Group on n letters.
A subgroup $A_n \overset{\text{def}}{=} \ker(\mathrm{Sgn})$ is called Alternating Group of degree n.
Clearly, $A_n \trianglelefteq S_n$ and $|S_n : A_n| = 2$, because $S_n / A_n \cong \{\pm 1\}$.

S_4.



Thm. S_4 is Solvable.

Proof. Since N_1 is subgroup of all elements of order 2, and Conjugate cannot change the order, $N_1 \trianglelefteq A_4$.

Now, $1 \overset{2}{\triangleleft} N_2 \overset{2}{\triangleleft} N_1 \overset{3}{\triangleleft} A_4 \overset{2}{\triangleleft} S_4$ all factors above are $\mathbb{Z}_2$ or $\mathbb{Z}_3$, thus abelian.

「Thm.

Let σ, τ be elements of the symmetric group S_n and suppose

$$\sigma = (a_1\ a_2 \cdots a_{k_1})(b_1\ b_2 \cdots b_{k_2}) \cdots$$

Then, the conjugate

$$\tau \circ \sigma \circ \tau^{-1} = (\tau(a_1)\ \tau(a_2) \cdots \tau(a_{k_1}))(\tau(b_1) \cdots \tau(b_{k_2})) \cdots$$

Proof. observe that if $\sigma(i) = j$, then

$$\tau \circ \sigma \circ \tau^{-1}(\tau(i)) = \tau(j)$$

「Def.

If $\sigma \in S_n$ is the product of disjoint cycles of lengths $n_1 \le n_2 \le \cdots \le n_r$, then the integers $n_1, \cdots, n_r$ are called the cycle type of σ.

「Prop.

Let $\sigma_1, \sigma_2 \in S_n$ be given. Then

σ_1 and σ_2 are conjugate if and only if σ_1 and σ_2 have same cycle type.

Further, the number of conjugacy classes of S_n equals the number of partitions of n.

Proof. If $\sigma_1 = \tau \sigma_2 \tau^{-1}$ for some τ, then the theorem above gives that they have same cycle structure. Conversely, suppose σ_1, σ_2 have same cycle type.

Thm. A_5 is simple group.

proof.